Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 3 W device runs for 5 s. Find transferred energy.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

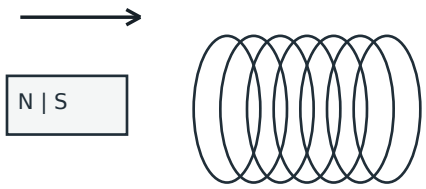
Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and mark the changing field through the loop. Contrast it with a stationary case.

Marzano check - circle: ① ② ③

03 Look and explain

Compare a magnet held still beside a fixed coil with a magnet moving into the coil. Which can produce a changing flux?



Coil

Marzano check - circle: ① ② ③

04 Build the solution

In a supplied trial, moving a magnet into a coil gives a positive meter deflection. Predict the sign when the same magnet is pulled out along the same path.

Marzano check - circle: ① ② ③

05 Use the idea

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Marzano check - circle: 1 2 3

06 Choose a method

Keep charge separation fixed and multiply one charge magnitude by 4. Compare electric-force magnitude.

Marzano check - circle: 1 2 3

07 Exit check

Name two ways to increase the magnitude of induced emf in a simple magnet-and-coil trial.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Predict how faster approach and more turns can change induced voltage. State which comparisons must keep the other conditions fixed.

Marzano check - circle: 1 2 3